BrandGuard AI — Round 2 Official Project Submission

CODEX 48 National Level Hackathon • 300 Teams Competing • Round 2 Qualification

Project: BrandGuard AI (Autonomous Brand Governance)	**Category:** Autonomous Systems / Enterprise AI
Team Leader: Priyansh Vaghela (priyansh@careplus.com)	**Core Member:** Sanskruti Antala (sanskruti@careplus.com)
Working Web Application: http://localhost:8000/app	**Authentication & Sign-in:** http://localhost:8000/login
Repository: Private Hackathon Monorepo (Git Tag: round2-submission)	**Database:** PostgreSQL 17.11 SSL (brandguard_db)

1. Executive Summary & Round 2 Qualification Objectives

BrandGuard AI is an enterprise pre-publication brand governance platform designed to eliminate multi-million dollar regulatory fines, trademark disputes, and misleading advertising violations before content goes live. Marketing creators submit promotional drafts, multi-channel ads, or packaging copy. The platform conducts autonomous real-time evaluations against strict corporate brand rulebooks and regulatory standards (ASCI, FTC, FDA), assigns deterministic risk scores, provides pinpoint explanations with compliant rewrite suggestions, and enforces a mandatory digital Human Sign-Off Gate restricted exclusively to Brand Managers.

2. List of Completed Features (Functional End-to-End Workflow)

The platform is fully operational from end to end. The judging panel can access the link and test all capabilities without setup:

- **Dual-Channel Review Studio:** Real-time evaluation of brand copy, promotional hashtags, and advertising claims against customizable regulatory rules.
- **Deterministic Multi-Tier AI Engine:** Powered by Google Gemini 1.5 with automated zero-latency regex fallback. Evaluates unconditional guarantees, illegal superlatives, discount caps (40% threshold), and mandatory topical disclaimers.
- **History & Past AI Answer Inspection Modal:** Complete transparency on past evaluations. Evaluators can click 'Inspect AI Answer' on any past record to view original copy, exact quoted ad text, detailed reasoning ('Why Wrong'), specific fix instructions ('Compliant Fix'), and copyable compliant AI suggested copy.
- **Mandatory Digital Sign-Off Gate:** High-risk campaigns cannot be published without formal digital sign-off. Only authenticated Brand Managers can approve drafts or request revisions; creator accounts are strictly restricted.
- **Role-Based User Isolation & Scoping:** Creators only access their individual draft history ('My Drafts'). Brand Managers possess an executive toggle to view workspace-wide submissions with creator attribution.
- **Enterprise Multi-User Authentication:** Complete registration and login system with bcrypt password hashing, persistent sessions, job title, company, employee ID badges, and workspace authorization codes.
- **Persistent PostgreSQL 17.11 Database:** Persistent storage with connection pooling for users, active brand guidelines, and immutable compliance audit trails.

3. Technology Stack & Implementation Architecture

Layer	Components & Technologies	Implementation Role
Frontend SPA	HTML5, CSS3, JavaScript ES6+, DM Sans & Manrope Typography	Executive telemetry, Studio workspace, Review Queue & Audit Modal
Backend API	FastAPI, Uvicorn ASGI, Pydantic v2, Python 3.14	RESTful endpoints, asynchronous evaluation pipeline & RBAC gating
Database	PostgreSQL 17.11 SSL (psycopg 3.2), SQLAlchemy 2.0	Relational audit storage, brand rules table & user credentials
AI Engine	Google Gemini 1.5 Flash (google-genai SDK), Fallback Rules	Natural language violation detection, citation & compliant rewriting
Security	BCrypt password hashing, session tokens, verification codes	Enterprise authentication & role permission enforcement

4. Changes and Enhancements from Round 1 Proposal

Based on preliminary testing and hackathon mentor feedback, the following engineering enhancements were incorporated:

- **PostgreSQL Database Upgrade:** Replaced lightweight in-memory dictionaries with a production-grade PostgreSQL 17.11 relational schema supporting multi-user concurrent access.
- **Direct Audit History Inspection:** Added a dedicated Inspection Modal to the Audit History view, enabling evaluators to inspect past AI findings, quoted violations, and compliant fixes without disrupting their active Studio session.
- **Stricter Digital Sign-Off Enforcement:** Upgraded the Sign-Off Gate so approval buttons are strictly restricted to authenticated Brand Managers when a draft is in the Review Queue.

5. Three Major Improvements Planned for Round 3 (Judging Weight: 15 Points)

For the National Finals (Round 3), the team will implement three substantial, enterprise-grade capabilities:

■ Improvement 1: Multimodal Computer Vision for Creative Asset Scanning (Figma, Canva & Ad Creatives)

Expands BrandGuard beyond text to analyze image and video creatives. Directly parses raster images, banners, and video frames via Gemini 1.5 Flash Vision to detect non-compliant logos, misleading visual fine-print, missing safety badge proportions, and competitor trademark infringements.

■ Improvement 2: Autonomous Headless CMS & Ad Network Gating Webhooks (Shopify, Meta & Google Ads)

Builds bi-directional webhook connectors for Shopify, Meta Marketing API, and Google Ads. BrandGuard acts as an autonomous pre-flight firewall: unapproved creatives are automatically intercepted and quarantined before public ad spend or store publishing can occur.

■ Improvement 3: Autonomous Regulatory Vector Knowledge Base (RAG on ASCI, FTC & FDA Guidelines)

Deploys pgvector semantic search over official advertising compliance frameworks: ASCI (Advertising Standards Council of India), US FTC Endorsement Guides, and FDA Cosmetic Claim Regulations. When copy is drafted, BrandGuard automatically cites relevant legal case law, penalty risks, and regulatory precedents in real time.

6. Usability & Self-Guided Evaluation (Judging Weight: 10 Points)

Judges can open the working project link and experience the entire platform without assistance:

- **Zero Setup:** All test accounts are pre-seeded in PostgreSQL with realistic brand compliance records.
- **Visual Status Cues:** High-contrast risk badges, dynamic approval timestamps, and contextual role warning banners guide the evaluator through each decision gate.
- **Responsive Layout:** Seamlessly adapts across laptop displays and mobile viewport sizes with collapsible navigation.